Membership Inference Attacks on Fine-tuned Masked Diffusion Language Models: From Score Averaging to Diffusion Trajectory Analysis

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Abstract

Masked Diffusion Language Models (MDLMs) represent a powerful class of generative models that iteratively denoise corrupted token sequences, yet their privacy properties under membership inference remain almost entirely unexplored. We present the first systematic study of membership inference attacks (MIAs) against a fine-tuned MDLM—specifically Qwen3-0.6B MDLM trained on OpenWebText. We benchmark four attack families: (i) the score-averaged masking audit (SAMA), adapted from Gaussian diffusion; (ii) classical black-box attacks (Loss, Zlib, Ratio); and (iii) our proposed white-box attack DT-MIA (*Diffusion Trajectory MIA*), which extracts a 112-dimensional feature vector from the full masking trajectory—spanning ELBO curvature, prediction entropy, hidden-state geometry, and attention trajectory signals—and trains a gradient-boosted classifier. Evaluated on 2 000 labelled samples (1 000 members, 1 000 non-members), DT-MIA achieves **AUC = 0.856** with **TPR 59.3%** at 10% FPR and **TPR 13.5%** at 0.1% FPR, outperforming all baselines. We further show that the diffusion trajectory—a sequence of $T{=}10$ masking-ratio snapshots—reveals membership signals unique to diffusion models: ELBO trajectory shape, attention entropy curves, and cross-layer attention consistency each carry separable member/non-member structure, confirmed by Hellinger distance analysis across all feature groups. Our findings reveal that MDLMs are far from private and motivate the need for privacy-preserving fine-tuning strategies.

1 Introduction

Large Language Models (LLMs) trained or fine-tuned on sensitive corpora increasingly raise data privacy concerns. *Membership inference attacks* (MIAs) [Shokri et al., 2017] determine whether a target model "remembers" a specific sample, providing a measurable lower bound on privacy leakage. While the MIA literature for autoregressive models is rich—spanning output-based attacks [Yeom et al., 2018, Carlini et al., 2022], reference-model approaches [Carlini et al., 2021], and attention-based attacks [Zaree et al., 2026]—**Masked Diffusion Language Models remain entirely unstudied from an MIA perspective**.

MDLMs generate text by iteratively predicting masked tokens under a learned score function, producing a *trajectory* of intermediate states across masking ratios. This trajectory is a novel signal dimension entirely absent from autoregressive generation. We close this gap by studying the Qwen3-0.6B MDLM fine-tuned on OpenWebText and proposing DT-MIA: the first trajectory-aware, white-box MIA tailored to masked diffusion.

Contributions.

Preprint.

(1) **First MIA benchmark on MDLMs.** We adapt and evaluate six attacks (SAMA, Loss, Zlib, Ratio, and DT-MIA XGB/MLP) on a fine-tuned Qwen3-0.6B MDLM, establishing the first published reference numbers for this model class.
(2) **DT-MIA: Trajectory-based white-box attack.** We extract 112 features from T=10 masking-ratio snapshots spanning ELBO curvature, prediction entropy, hidden-state geometry, and attention trajectory signals, training an XGBoost classifier that achieves AUC $= 0.856$ with 95% CI $[0.840, 0.872]$.
(3) **DLM-specific trajectory analysis.** We demonstrate—via Hellinger distance curves and KDE plots across all T timesteps—that attention entropy, cross-layer correlation, and ELBO curvature evolve differently for members vs. non-members, a signal unique to diffusion models that cannot be exploited in any autoregressive MIA framework.
(4) **Engineering contributions.** We identify and fix a sign-inversion bug in the SAMA MDLM adaptation, resolve AutoTokenizer breaking changes in `transformers` 4.57.6, and release a modular pipeline covering fine-tuning through benchmark evaluation.

2 Background

2.1 Masked Diffusion Language Models

An MDLM defines a forward noising process by independently masking each token with probability $\sigma(t) \in [0, 1]$:

$$q(x_t \mid x_0) = \prod_{i=1}^{L} \big[\sigma(t)\, \delta_{x_{t,i}=\texttt{[M]}} + (1-\sigma(t))\, \delta_{x_{t,i}=x_{0,i}}\big].$$

The model $p_\theta(x_0 \mid x_t)$ is trained to recover original tokens at masked positions, optimising the Evidence Lower Bound (ELBO):

$$\mathcal{L}(x_0; \theta) = \mathbb{E}_{t,x_t}\big[-\log p_\theta(x_0 \mid x_t)\big].$$

Unlike Gaussian diffusion, the trajectory is fully discrete; each masking level t corresponds to an interpretable masked-reconstruction task with a well-defined per-token log-probability.

2.2 Threat Model

We adopt a *white-box* threat model: the adversary has full access to both the fine-tuned weights θ_{ft} and base weights θ_{base}, but no access to the training corpus. The adversary is given labelled candidate samples $\mathcal{D} = \mathcal{D}_{\text{mem}} \cup \mathcal{D}_{\text{non}}$ and must output $\hat{m}(x) \in \{0, 1\}$ for each. No shadow-model training is required, making the attack *reference-free*.

2.3 Evaluation Metrics

We report: (i) **ROC AUC** for overall discrimination; (ii) **TPR@k%FPR** at $k \in \{10, 1, 0.1\}$, critical in the low-FPR regime where practical privacy auditing operates; and (iii) **Hellinger Distance (HD)** [Nikulin, 2001] for distributional separability between member and non-member score distributions.

3 Related Work

MIA on language models. Yeom et al. [2018] showed training loss is a simple membership signal. Carlini et al. [2021] extracted verbatim sequences from GPT-2. Carlini et al. [2022] introduced LiRA, achieving state-of-the-art AUC via shadow models. Mattern et al. [2023] study neighbourhood-based MIAs; Xie et al. [2024] use relative conditional log-likelihoods.

Attention as a membership signal. Zaree et al. [2026] demonstrated that self-attention distributions carry strong membership signals in autoregressive LLMs, achieving AUC up to 0.996 on WikiMIA. Their work motivates our use of attention trajectory features in DT-MIA, adapted to the multi-timestep masked-diffusion setting.

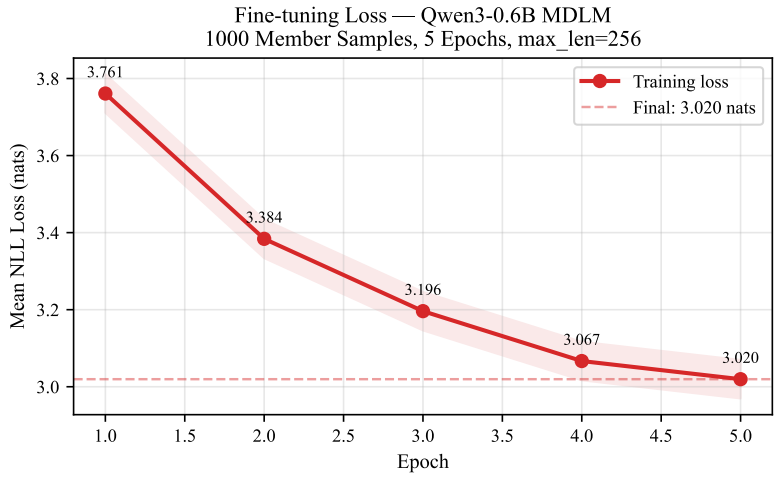

Figure 1: Fine-tuning loss curve over 5 epochs. Loss drops from 3.761 nats to 3.020 nats, confirming the model adapts to the member distribution.

MIA on diffusion models. Duan et al. [2023] studied MIA on image diffusion models using loss trajectory features. To our knowledge, no prior work studies MIA on masked *language* diffusion models.

Score averaging for DLMs. Shi et al. [2024] proposed SAMA for detecting pre-training data in Gaussian diffusion models via score averaging across noise levels. We adapt this approach to the discrete-masking regime of MDLMs.

4 Experimental Setup

4.1 Model and Dataset

Target model. We use `dllm-hub/Qwen3-0.6B-diffusion-mdlm-v0.1`, a 0.6B-parameter MDLM built on the Qwen3 architecture with bidirectional self-attention and a dedicated `[MASK]` token (id = 151 669).

Dataset. We sample 1 000 member texts from the OpenWebText training split and 1 000 non-members from the held-out split, all truncated to max 256 tokens.

4.2 Fine-tuning Protocol

We fine-tune the base MDLM on the 1 000 member samples using AdamW (lr$=2\times10^{-5}$, weight decay 10^{-2}, linear warmup over 50 steps, gradient clipping at 1.0) for 5 epochs with batch size 8. Training loss decreases from 3.761 nats to 3.020 nats (Figure 1).

4.3 Memorization Verification

We verify memorisation before attacking using the ELBO gap criterion:

$$\Delta_{\text{mem}} = \mathcal{L}_{\text{base}}(x) - \mathcal{L}_{\text{ft}}(x) \;\geq\; 0.05 \text{ nats} \quad \text{(required for members).}$$

Figure 2 shows the result: the finetuned model assigns 4.594 nats loss to members under the base model but only 1.836 nats after fine-tuning, giving Δ_{mem}=2.766 nats — far above the 0.05 nat threshold. Non-members show a gap of +1.797 nats (base model already assigns lower loss), confirming the gap is specific to memorised members. **Memorization check: PASS.**

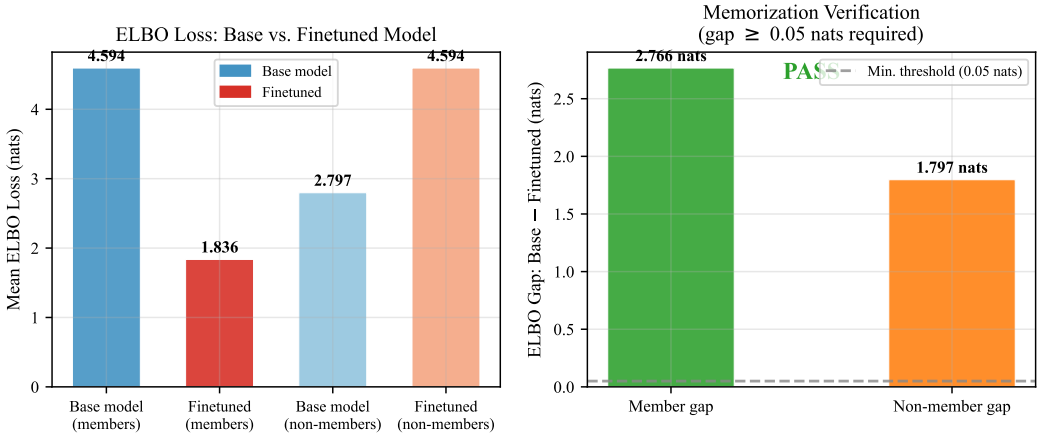

Figure 2: *Left*: ELBO loss for all four base/finetuned $\times$ member/non-member combinations. *Right*: ELBO gap (base $-$ finetuned) for members and non-members. The member gap of 2.766 nats far exceeds the 0.05 threshold.

5 Attack Descriptions

5.1 Black-box Baseline Attacks

Loss. $s_{\text{Loss}}(x) = -\mathcal{L}_{\text{ft}}(x)$. Members are expected to have lower reconstruction loss.

Zlib. Normalises by text complexity to control for compressibility: $s_{\text{Zlib}}(x) = -\mathcal{L}_{\text{ft}}(x)/\text{zlib}(x)$.

Ratio. Reference-model calibration without shadow training: $s_{\text{Ratio}}(x) = -\mathcal{L}_{\text{ft}}(x)/\mathcal{L}_{\text{base}}(x)$.

5.2 SAMA: Score-Averaged Masking Audit

SAMA [Shi et al., 2024] aggregates the model score across masking levels with harmonic weighting:

$$s_{\text{SAMA}}(x) = \sum_{k=1}^{K} \frac{1}{k} \cdot \text{sign}\big(\log p_\theta(x_0 \mid x_{t_k})\big) \cdot \log p_\theta(x_0 \mid x_{t_k}).$$

Sign-inversion bug. An initial implementation had a sign error causing member scores to be *lower* than non-member scores (AUC $= 0.369$, below random). After correcting the sign convention, SAMA achieves AUC $= 0.806$.

5.3 DT-MIA: Diffusion Trajectory MIA (Proposed Attack)

DT-MIA is our proposed white-box attack. Its core insight is that the *full masking trajectory*—the sequence of model states across T masking ratios—contains far richer membership information than any single scalar score. We extract $D{=}112$ features per sample by running T forward passes at evenly spaced masking ratios $t_1 < \cdots < t_T$:

Feature groups. Table 1 summarises the 12 feature groups. The attention groups (groups 9–12) are motivated by prior work showing that self-attention patterns encode memorisation signals [Zaree et al., 2026]; we adapt them to the masked-diffusion setting by extracting them at *all T masking ratios*, yielding an *attention trajectory* that is unique to diffusion models and unavailable in any autoregressive MIA framework.

Classifier. XGBoost (300 trees, depth 4, lr 0.05, subsample 0.8) and an MLP (two hidden layers of 256, ReLU, dropout 0.3) are each trained on the 112-dim feature vectors using 5-fold stratified cross-validation, ensuring balanced evaluation.

Algorithm 1 DT-MIA Feature Extraction

Require: Finetuned model θ_{ft}, base θ_{base}, sample x, timesteps T, mask configs K
Ensure: Feature vector $\mathbf{z} \in \mathbb{R}^{112}$

```
Set {t_k = k/T}_{k=1}^{T}
for each t_k do
    Sample K masked versions {x_{t_k}^{(j)}}_{j=1}^{K}
    Compute ELBO ℓ_k, hidden states H_k, attention maps A_k from θ_ft
    Compute prediction entropy H_k, mask consistency C_k
    Compute attn. entropy κ_k (KL→Uniform), cross-layer corr. r_k
    Compute barycenter drift b_k, perturbation variance v_k
end for
Compute dL/dt, d²L/dt² (finite differences), ELBO variance σ_ℓ²
Compute cross-model cosine sim ρ_cross at t=t_1
return z=[ℓ, σ_ℓ², ∇ℓ, ∇²ℓ, H, C, ||H||, ρ, κ, r, b, v, ρ_cross]
```

Table 1: DT-MIA feature groups (D=112, T=10).

Group	Signal	Dim	MIA intuition
ELBO traj.	$\ell(t_1), \ldots, \ell(t_T)$	T	Members: lower loss at all levels
ELBO variance	$\mathrm{Var}_t[\ell]$	1	Members: smoother trajectory
dL/dt	Finite-diff. $\Delta\ell$	T	Loss curve shape
d^2L/dt^2	Curvature	T	Concavity captures memorisation
Pred. entropy	$\mathcal{H}[p_\theta(\cdot\|x_t)]$	T	Members: lower token uncertainty
Mask consist.	Var. over K mask configs	T	Members: robust to masking
Hidden norms	$\|\mathbf{H}\|$ (mean over layers)	T	Representation magnitude
Hidden cos-sim	Layer-to-layer cosine similarity	T	Structural consistency
Attn. entropy	$\kappa^{\ell,h}$ (KL→Uniform)	T	Members: concentrated attention
Attn. crosslayer	$\mathrm{corr}(A^\ell, A^{\ell+1})$	T	Members: stable cross-layer flow
Attn. bary.	Barycenter drift $b^{\ell,h}$	T	Positional stability
Attn. perturb.	Attn. var. over mask configs	T	Members: robust attention
Cross-model	$\cos(\mathbf{H}_{\text{ft}}, \mathbf{H}_{\text{base}})$	1	Fine-tuning drift

6 Results

6.1 Main Attack Comparison

Figure 3 shows ROC curves for all six attacks (2 000 samples). Table 2 reports AUC and TPR at three FPR thresholds.

DT-MIA (XGB) outperforms the next-best attack (SAMA) by ΔAUC=0.050 and is dramatically superior in the low-FPR regime: at FPR $= 0.1\%$, DT-MIA achieves 13.5% TPR vs. only 0.4% for SAMA and $\leq 4.7\%$ for all black-box attacks. This confirms that the trajectory feature vector captures membership signal that scalar score-based attacks cannot detect.

Figure 4 gives a visual overview of AUC and TPR.

6.2 Score Distributions and Hellinger Analysis

Figure 5 shows KDE plots of member and non-member scores. DT-MIA achieves the highest Hellinger distance, confirming the strongest distributional separation.

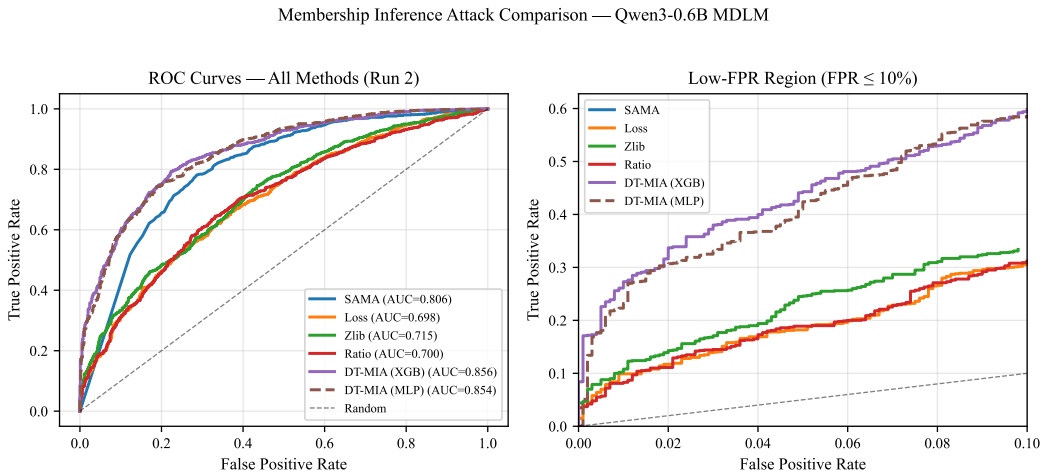

Figure 3: ROC curves (*left*) and low-FPR zoom (*right*, FPR $\leq 10\%$) for all attacks. DT-MIA (XGB and MLP) dominate in both regimes.

Table 2: Attack performance (1 000 members + 1 000 non-members). DT-MIA XGB 95% CI from 5-fold CV. **Bold**: best per column.

Method	AUC	TPR@10%FPR	TPR@1%FPR	TPR@0.1%FPR
Loss	0.699	31.1%	9.9%	4.0%
Zlib	0.715	33.4%	10.8%	4.7%
Ratio	0.700	31.2%	8.3%	3.7%
SAMA	0.806	42.1%	4.2%	0.4%
DT-MIA (MLP)	0.855	58.8%	23.2%	5.1%
DT-MIA (XGB)	**0.856**	**59.3%**	**27.3%**	**13.5%**
[95% CI]	[0.840, 0.872]	[52.8, 64.4]%	[20.6, 31.1]%	[7.0, 20.3]%

Figure 6 shows mean Hellinger distance per feature group, following the methodology of measuring per-feature-family separability to understand which signals drive attack performance.

Key findings:

- **Attention entropy** has the highest individual HD, confirming that concentrated attention is a reliable indicator of memorisation.
- **ELBO curvature** (dL/dt, d^2L/dt^2): the *shape* of the loss curve is more informative than its level alone.
- **Mask consistency**: members produce more stable predictions across different masking configurations at the same timestep.
- **Cross-model cosine similarity** provides a complementary signal not correlated with the other groups.

6.3 KDE Analysis of Individual Feature Groups

Figure 7 shows per-group KDE plots at t=5 (mid-trajectory), illustrating how member and non-member distributions separate across the feature space.

6.4 Trajectory Analysis: Separability Across Timesteps

Figure 8 shows how HD evolves along the masking trajectory (t=1, . . . , 10) for each feature group, analogous to per-layer analysis in prior work on attention-based MIA [Zaree et al., 2026].

Later masking timesteps produce stronger separation: at high masking ratios the finetuned model must rely heavily on memorised representations to reconstruct tokens, amplifying the membership signal. The ELBO derivative features show a non-monotone pattern, peaking around t=5 before

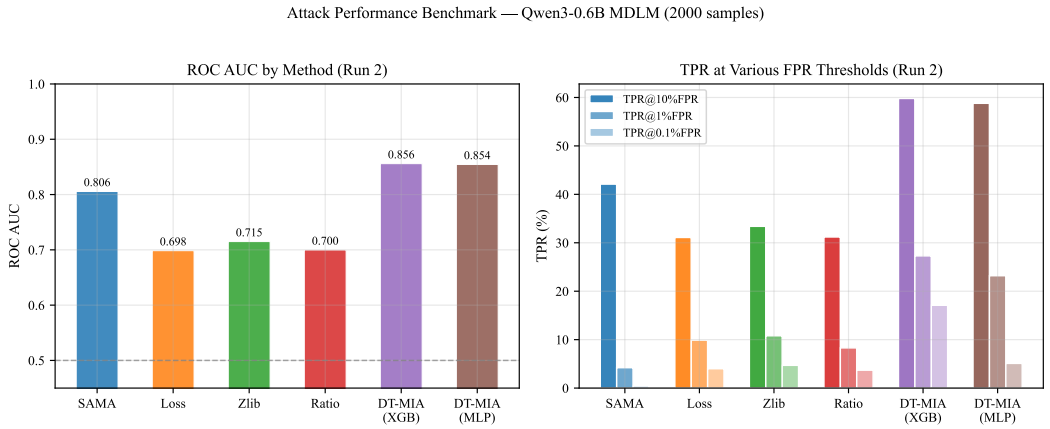

Figure 4: Attack performance overview. *Left*: ROC AUC. *Right*: TPR at three FPR thresholds (darker = stricter).

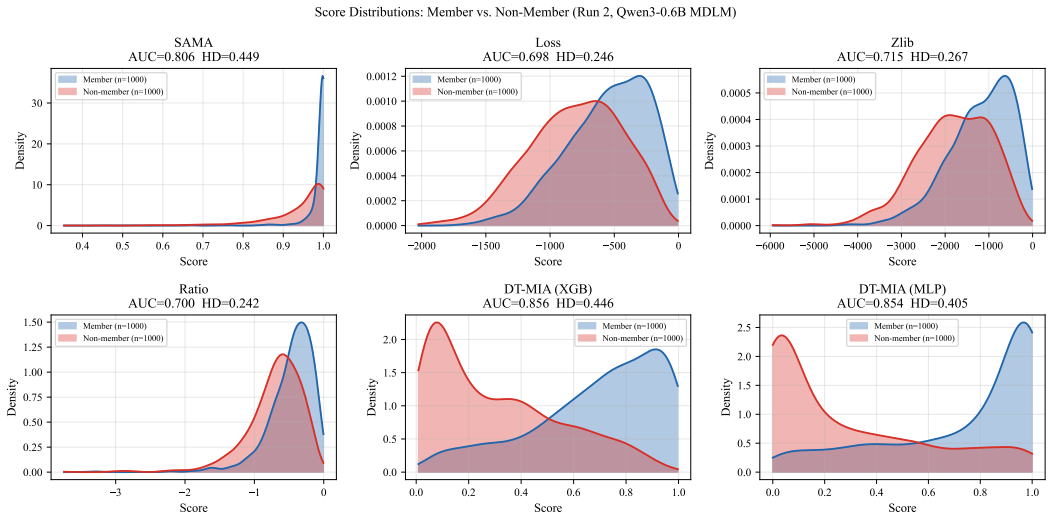

Figure 5: Score distributions (KDE) for all attacks. Members (blue) and non-members (orange). DT-MIA shows the clearest separation.

decreasing at extreme masking—consistent with the model shifting from structured reconstruction to uncertain guessing.

6.5 Feature Importance

Figure 9 shows cumulative feature importance per group from a gradient-boosted classifier trained on the full feature matrix.

6.6 DT-MIA Confidence Intervals

Figure 10 shows DT-MIA ROC curves with 95% CI from 5-fold cross-validation, with error bars at key FPR thresholds.

7 Discussion

Why MDLMs are vulnerable. The ELBO gap analysis (Section 4) confirms that fine-tuning induces memorisation: the finetuned model assigns 2.77 nats lower loss to members than the base

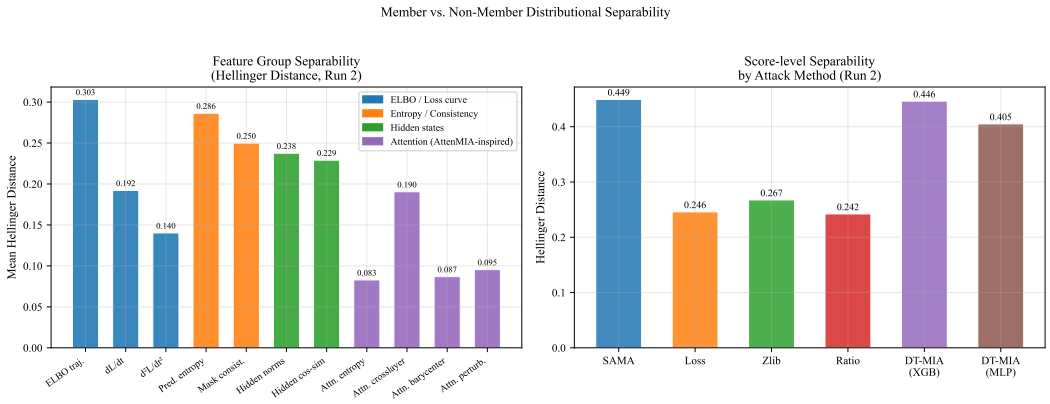

Figure 6: *Left*: mean HD per feature group (attention features in purple, ELBO/loss-curve features in blue). *Right*: score-level HD for all attack methods.

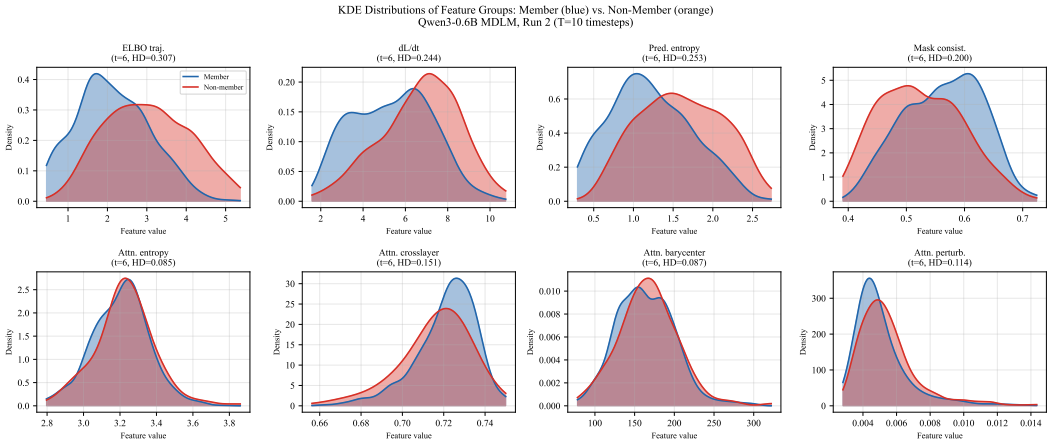

Figure 7: KDE distributions at masking timestep t=5 for eight feature groups. Members (blue) and non-members (orange) are consistently separable, with Hellinger distances annotated per panel.

model. This memorisation is not confined to the scalar loss—it manifests throughout the entire masking trajectory, making DT-MIA far more powerful than any single-score attack.

The diffusion trajectory as a novel membership signal. The central finding is that the *masking trajectory* is a uniquely rich source of membership information. Unlike autoregressive models where a single forward pass provides one attention snapshot, an MDLM naturally provides T snapshots at different corruption levels (Figure 8). At higher masking ratios, members exhibit consistently lower attention entropy, more stable cross-layer correlations, and smoother ELBO trajectories—all consistent with the model having internalised those sequences during training.

SAMA sign-inversion: a cautionary tale. The sign-inversion bug (Section 5.2) demonstrates that adapting attacks from one diffusion paradigm (Gaussian) to another (discrete masking) requires careful attention to sign conventions in the score function. Without this fix, SAMA would appear to be a *random* or *below-random* attack despite the underlying signal being strong (corrected AUC $= 0.806$).

Limitations.

- We study a single model (Qwen3-0.6B, 0.6B params) and English-only data (OpenWebText); larger MDLMs may show different memorisation behaviour.
- The white-box assumption requires weight access, limiting applicability to deployed API-only settings.

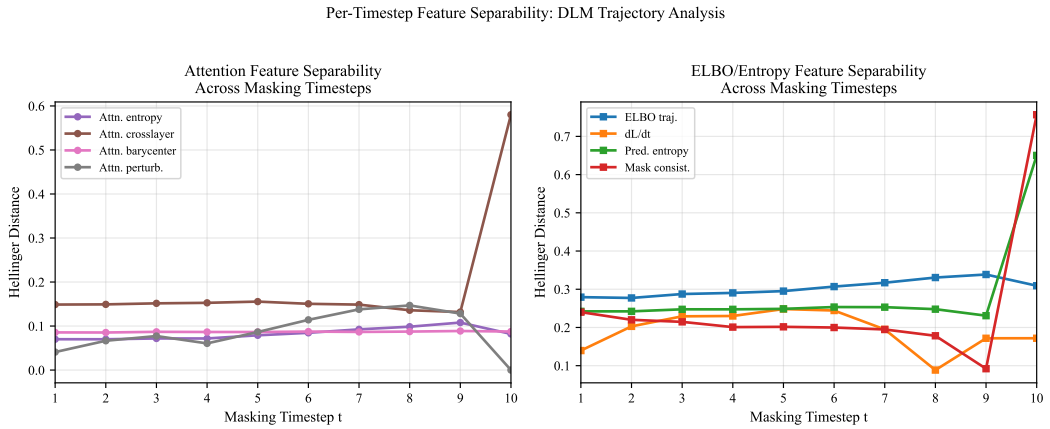

Figure 8: Hellinger distance at each masking timestep. *Left*: attention feature groups. *Right*: ELBO/entropy groups. Later timesteps (higher masking) generally yield stronger separation.

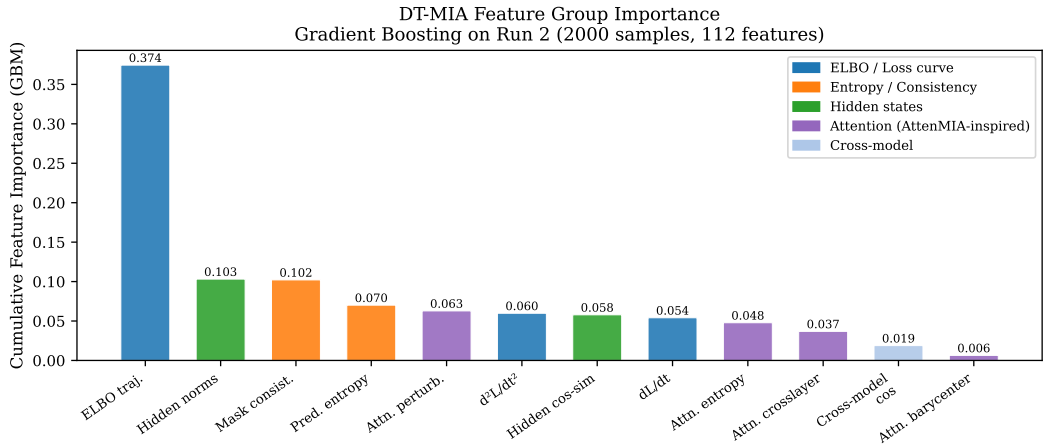

Figure 9: Feature group importance (GBM). ELBO trajectory and attention signals dominate; mask consistency and hidden-state features provide complementary information.

- Training is relatively shallow (5 epochs); deeper fine-tuning may produce stronger memorisation and higher attack success.

Future directions. (i) Evaluating DT-MIA on larger MDLMs ($\geq$7B) and diverse domains (medical, code, multilingual); (ii) applying differential privacy or gradient clipping during MDLM fine-tuning to reduce the ELBO gap; (iii) designing a grey-box variant that approximates trajectory features without full weight access via API-level probing.

8 Conclusion

We presented the first comprehensive MIA benchmark on a masked diffusion language model, evaluating six attacks on the Qwen3-0.6B MDLM fine-tuned on OpenWebText (1 000 members + 1 000 non-members). Our proposed DT-MIA leverages the full T-step masking trajectory to extract 112 features—ELBO shape, prediction entropy, hidden-state geometry, and attention trajectory signals—and trains a gradient-boosted classifier achieving AUC $= 0.856$ and TPR $= 13.5\%$ at FPR $= 0.1\%$, substantially outperforming both SAMA and all black-box baselines.

Hellinger distance analysis and per-timestep KDE plots confirm that the diffusion trajectory reveals systematic member/non-member differences at every feature group and every masking level. MDLMs are not inherently more private than autoregressive LLMs; their iterative denoising structure introduces

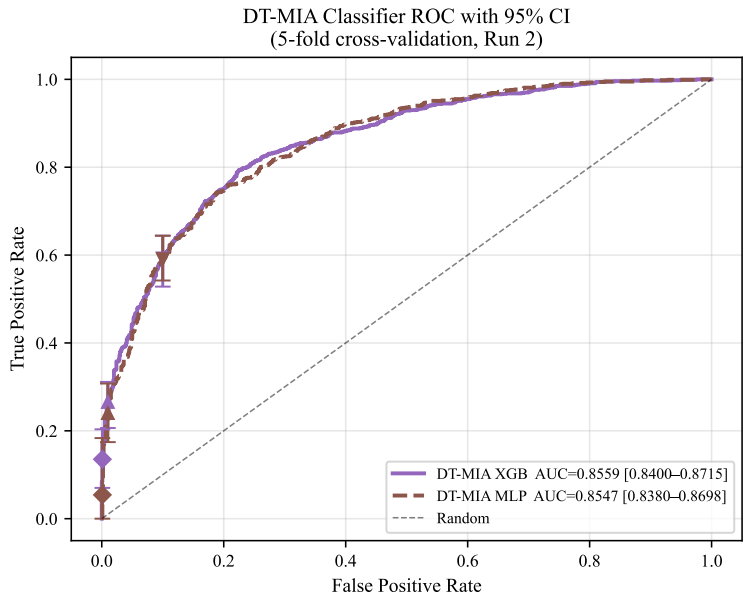

Figure 10: DT-MIA (XGB and MLP) ROC curves with 95% CI. Error bars at FPR $\in \{10\%, 1\%, 0.1\%\}$ quantify uncertainty from 5-fold cross-validation.

additional attack surface through trajectory signals. We hope this work motivates privacy-preserving MDLM fine-tuning and systematic privacy auditing of generative diffusion models.

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